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DIGITAL FORENSICS FUNDAMENTALS:

Forensics is the application of methods and procedures to investigate and solve crimes. The branch of forensics associated with cyber crime is known as digital forensics. Some procedures are followed by digital forensics team while collecting, storing, analyzing and reporting the evidence.

- Four phases of process of digital forensics:

1= Collection: In this phase we collect data from devices such as laptops, digital cameras, USB's etc found on the crime scene. the original data should not be harmed while collecting the evidence.

2= Examination: In this phase we need to filter the data and the data which is needed is extracted. Then the filtered or the extracted data is moved to next phase.

3= Analysis: In this phase the data is analyzed by investigators by correlating multiple pieces of data to draw conclusions and aim to extract activities relevant to the case.

4= Reporting: In this phase a detailed report of the analysis performed by investigators is made containing investigation's methodology and detailed findings from the collected evidence.

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- As we can find different pieces of evidence from the crime scene. Analyzing these multiple categories of evidence requires various tools and techniques.

There are different types of digital forensics, all with their own collection and analysis methodologies.

1. Computer forensics: The most common is computer forensics which requires investigating computers, the devices most commonly used in crime.

2. Mobile forensics: It involves investigating mobile devices found on crime scene. In this, data such as call records, text messages, GPS location and more are extracted.

3. Network forensics: In this the whole network is investigated. The majority evidence found in this are network traffic logs.

4. Database forensics: Database forensics involves investigation of any intrusion into database that resulted in data modification.

5. Cloud forensics: It involves investigation of data stored in cloud infrastructure.

6. Emails forensics: In this forensics the emails are investigated that whether they are a part of phising or fraudlent campaigns.

- Evidence Acquisition: Evidence acquisition methods for digital devices depends on type of digital device. Some practices that must be followed while acquiring evidence.

1. Proper Authorization: Proper authorization from the relevant authorities must be obtained before collecting evidences from

Write blocker : It is a tool that allows you to read data from a device but prevents any changes to it.

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the crime scenes because it may contain individuals or any organization's sensitive data.

b) Chain of custody : Chain of custody is a document containing all the details of the evidence and records who handled the evidence. It can be used to prove the integrity and reliability of the evidence in court.

c) Use of write blocker : write blockers must be used during any forensic team task because without it the timestamps of the files on the hard drive may alter which are the important evidence of crime.

- Windows forensics : It is the investigation of windows operating system. During the data collection phase forensic images of the windows OS is taken. Forensic images are the bit by bit copy of the whole OS.

There are two different catogaries of forensic image taken :

1 = Disk image : The disk image contains all the data present in the system (HDD, SSD, etc). this is non-volatile means it would survive even after a restart of OS.

2 = Memory image : Contains data inside the OS ram. For example running processes, current network connection, open files, etc It should be taken first because it is volatile and gets deleted after OS is shutdown.

Some tools for disk and memory image acquisition and analysis of windows OS are :

• Metadata: Hidden information about a file. Ex: in an image - Camera model, Date & Time, GPS location, Author, Software used.

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a) FTK Imager: It is a tool used for disk image of windows OS and can also be used for analysis purposes.

b) Autopsy: It is a popular open source digital forensics platform where investigator can import disk image and this tool will conduct an extensive analysis of the image.

c) Dumpit: This tool is used to take memory image of the windows OS using a command-line interface and few commands.

d) Volatility: It is a tool used to analyze memory images.

• Exiftool: It is a tool used to read and edit metadata of a file.
Command - exiftool filename
• Supports: Images, videos, Audio, pdf, documents

pdfinfo: It is a tool used to extract basic metadata for pdf files only.

→ It extracts detailed metadata, not just basic.